

YEAR 11, HSC MATHEMATICS - ASSESSMENT 1

Marks

Question 1 (13 marks)

- (a) Determine a quadratic equation in general form with roots equal to 2 and -1. **2**
- (b) Differentiate each of the following
- i) $\sqrt{6x-5}$ **2**
- ii) $3x(2x-1)^2$ **3**
- (c) If α and β are the roots of the equation $2x^2 + 3x + 5 = 0$, find the value of
- i) $\alpha + \beta$ **1**
- ii) $\alpha\beta$ **1**
- iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ **2**
- iv) $\alpha^2 + \beta^2$ **2**

Question 2 (14 marks)

(Start a new page)

- (a) Consider the parabola with equation $x^2 = -8(y + 3)$ 4
- i) Find the coordinates of the vertex of the parabola.
- ii) Find the coordinates of the focus of the parabola.
- (b) Solve the quadratic inequality $2x^2 - x - 10 \leq 0$ 3
- (c) Find two numerical values of p so that $x^2 + 2px + (7p + 8) = 0$ has equal roots. 4
- (d) Solve the equation $(x^2 - x) - 18(x^2 - x) + 72 = 0$ 3

Question 3 (15 marks)

(Start a new page)

- (a) Find the value of m if one root of the equation $x^2 + 6x + m = 0$ is double the other. 3
- (b) Find the values of A , B , and C if $x^2 - x \equiv A(x - 4)^2 + B(x - 4) + C$. 4
- (c) Let A and B be the fixed points $(-1, 0)$ and $(2, 0)$ and let P be the variable point (x, y) . 8
- (i) Write down expressions for the distances PA^2 and PB^2
- (ii) Suppose that P moves so that $PA = 2PB$. Show that the equation of this locus of points is the circle $x^2 - 6x + y^2 + 5 = 0$
- (iii) Hence find the centre and radius of the circle.

End of paper

Q1 a) $(x-2)(x+1) = x^2 - x - 2 = 0$

b) i) $\frac{d}{dx} (6x-5)^{\frac{1}{2}} = \frac{1}{2} (6x-5)^{-\frac{1}{2}} \times 6$

ii) $\frac{d}{dx} 3x(2x-1)^7 = 3(2x-1)^7 + 3x \times 7(2x-1)^6$
 $= 3(2x-1)^7 + 21x(2x-1)^6$

iii) $\frac{d}{dx} \frac{x}{2x-3} = \frac{2x - (2x-3)x}{(2x-3)^2}$
 $= \frac{3}{(2x-3)^2}$

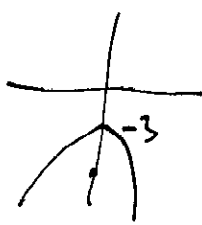
c) i) $\alpha + \beta = \frac{-b}{a} = -\frac{3}{2}$

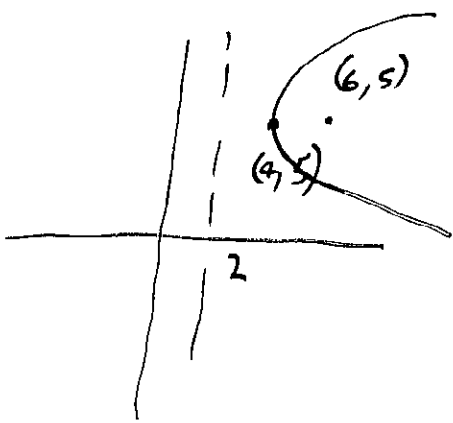
ii) $\alpha\beta = \frac{c}{a} = \frac{5}{2}$

iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$
 $= \frac{-3/2}{5/2}$
 $= -\frac{3}{2} \times \frac{2}{5}$
 $= -\frac{3}{5}$

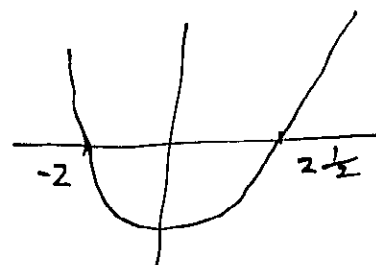
iv) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (-\frac{3}{2})^2 - 2 \times \frac{5}{2}$
 $= \frac{9}{4} - 5$
 $= -\frac{11}{4}$

Q2

- a)  i) vertex $(0, -3)$ ✓
 ii) $a = 2$ ✓
 focus $(0, -5)$ ✓

- b)  $a = 2$
 $(y - 5)^2 = 8(x - 4)$
 $a = 2$ ✓
 vertex $(4, 5)$ ✓
 $(y - k)^2 = 4a(x - h)$ ✓

- c) $2x^2 - x - 10 \leq 0$
 $p = -20$
 $q = -1$
 $F = -5, 4$
 $2x^2 - 5x + 4x - 10 \leq 0$
 $x(2x - 5) + 2(2x - 5) \leq 0$
 $(2x - 5)(x + 2) \leq 0$
 $-2 \leq x \leq 2\frac{1}{2}$ ✓



2 marks for factorising,
 1 mark for answer.

- d) ~~let $u = 5^x$~~
 ~~$u^2 - 6u + 5 = 0$~~
 ~~$(u - 1)(u - 5) = 0$ ✓~~
 ~~$u = 1, u = 5$ ✓~~
 ~~$5^x = 1 \quad 5^x = 5$~~
 ~~$x = 0 \quad x = 1$ ✓~~

Question 3

a) let $u = 5^x$

$$u^2 - 6u + 5 = 0$$

$$(u-1)(u-5) = 0 \quad \checkmark$$

$$u=1, u=5 \quad \checkmark$$

$$5^x = 1$$

$$x=0$$

$$5^x = 5$$

$$x=1 \quad \checkmark$$

b) let the root be d & $2d$

$$d + 2d = \frac{-6}{1} \quad \checkmark$$

$$3d = -6$$

$$d = -2 \quad \checkmark$$

$$d \times 2d = \frac{m}{1}$$

$$-2 \times -4 = m$$

$$m = 8 \quad \checkmark$$

c) $A = b^2 - 4ac = 0$

$$(2p)^2 - 4(7p+8) = 0 \quad \checkmark$$

$$4p^2 - 28p - 32 = 0$$

$$p^2 - 7p - 8 = 0 \quad \checkmark$$

$$(p-8)(p+1) = 0$$

$$p = 8 \text{ or } -1 \quad \checkmark$$

Q4 a) $x^2 - x \equiv A(x-4)^2 + B(x-4) + C$

$$x^2 = Ax^2 \quad A=1$$

$$x^2 - x \equiv (x-4)^2 + B(x-4) + C$$

Let $x=4$

$$16-4 = 0 + 0 + C$$

$$C = 12$$

$$x^2 - x \equiv (x-4)^2 + B(x-4) + 12$$

Let $x=0$

$$0 \equiv 16 - 4B + 12$$

$$4B = 28$$

$$B = 7$$

$$A = 1, B = 7, C = 12$$

$$x^2 - x = (x-4)^2 + B(x-4) + C$$

$$= x^2 - 8x + 16 + Bx - 4B + C$$

$$= x^2 - 8x + Bx + 16 - 4B + C$$

$$= x^2 + x(B-8) + 16 - 4B + C$$

$$\therefore B-8 = -1$$

$$\underline{B=7} \quad A=1$$

$$\underline{\underline{\therefore A=1, B=7, C=12}}$$

$$16 - 4B + C = 0$$

$$16 - 4 \times 7 + C = 0$$

$$16 - 28 + C = 0$$

$$\underline{\underline{C=12}}$$

Q4 cont

$$b) i) PA = 2PB$$

$$\sqrt{(x+1)^2 + (y-0)^2} = 2\sqrt{(x-2)^2 + (y-0)^2} \quad \checkmark \checkmark$$

$$(x+1)^2 + y^2 = 4((x-2)^2 + y^2)$$

$$0 = 4(x-2)^2 - (x+1)^2 + 3y^2$$

$$0 = 4(x^2 - 4x + 4) - (x^2 + 2x + 1) + 3y^2 \quad \checkmark$$

$$0 = 3x^2 - 18x + 15 + 3y^2$$

$$0 = x^2 - 6x + 5 + y^2 \quad \checkmark$$

$$0 = x^2 - 6x + y^2 + 5$$

ii)

$$0 = x^2 - 6x + 9 + y^2 + 5 - 9$$

$$0 = (x-3)^2 + y^2 - 4 \quad /$$

$$(x-3)^2 + y^2 = 4$$

$$\text{centre } (3, 0) \quad \checkmark$$

$$\text{radius } 2 \quad \checkmark$$